Governance Incentive Engineering

A Testable Framework for Reality-Coupled Constitutional Systems Under Adversarial Optimization Pressure

Constitutional rights and admissibility remain prior constraints; they are not reducible to incentive optimization.

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

Declared rules do not determine behavior by themselves. Actors encounter permissions, prohibitions, evidence requirements, delays, sanctions, rights, technical affordances, institutional dependencies, and uneven capacities as a structured field of possible action. The source paper names the deliberate analysis and redesign of that field **Governance Incentive Engineering (GIE)**. Its core insight is that constitutional governance must examine not only what a system says is lawful, but which strategies become viable, stable, costly, correctable, observable, or dominant in operation.

This revision turns that insight into a bounded research framework. A governed system is represented as an actor-relative strategic transition system with separate constitutional, admissibility, observation, effect, payoff, cost, and revision components. Costs are vector-valued and distributional rather than a single universal energy. Binding cost, correction cost, and deception cost remain the source framework's central diagnostic dimensions, but they are joined by audit, participation, contest, exit, privacy, coordination, and harm-displacement costs. No scalar ranking is valid unless its actor, horizon, aggregation rule, uncertainty, and authority are declared.

The paper also narrows its strongest claim. Raising the tested cost of selected deceptive strategies can make those strategies less attractive within a specified model and observation regime. It cannot establish honest intention, objective truth, legitimate authority, universal equilibrium, or adversarial convergence. Actors may mimic compliance, capture witnesses, shift burdens, exit, sabotage, collude, exploit unobserved channels, or change the game. Constitutional rights and admissibility remain prior constraints; they are not reducible to incentive optimization.

GIE is therefore best understood as a candidate discipline for designing and testing the strategic consequences of governed arrangements while preserving contest,

refusal, privacy, pluralism, and revision. Semantic Manifold Topology supplies a way to represent thick relations and reachable continuations; Governance-Aware Optimization supplies a specific update-space realization; Constitutional Epistemology supplies the adversarial threat model; Admissibility Engineering supplies constraint-surface discipline; and the Governed Meaning Base can retain models, assumptions, disputes, experiments, and revisions without becoming their truth or authority. The resulting program is conceptual and testable, but not yet externally researched, empirically validated, or proven at civilization scale.

1. Problem, Thesis, and Claim Ceiling

Governance has at least two surfaces:

1. a **declared surface** of rules, authorities, rights, procedures, and reasons;
2. an **operational surface** of available actions, observed consequences, burdens, payoffs, delays, dependencies, and enforcement.

The two surfaces can diverge. A system may prohibit deception while rewarding compliant appearance. It may permit correction while making correction inaccessible. It may promise participation while concentrating agenda-setting, evidence production, and implementation authority. Conversely, a system can make some harmful strategies difficult to sustain even when it cannot observe intentions directly.

Governance Incentive Engineering is the candidate discipline of representing, testing, and revising the strategic consequences induced by a governed arrangement under declared constitutional constraints.

Its thesis is:

> A governance design is incomplete until its declared rules are evaluated against the actor-relative strategy landscape they produce, including who bears each cost, which conduct remains observable, which rights constrain optimization, and how error, contest, exit, and revision remain possible.

This paper does not establish that:

- governance is reducible to incentives;
- all actors maximize a stable utility function;
- social or semantic systems possess literal physical energy landscapes;
- observed compliance proves honesty, truth, consent, legitimacy, or justice;
- one cost metric can compare all actors or values;
- a designer can know the complete strategy space;
- adversarial convergence to honest behavior is generally achievable;
- every governance pathology is a topological failure;
- the named metrics are operationally validated;
- GIE is externally novel or distinct from every adjacent discipline;
- any institutional or technical implementation is authorized.

The current claim is narrower: a typed model of rules, transitions, observations, effects, and actor-relative costs can expose strategic failure modes that rule inspection alone misses, and controlled interventions can test whether selected harmful strategies become less viable without violating prior constitutional obligations.

2. Constitution Is Not an Incentive Function

GIE must preserve several distinctions that optimization language tends to collapse.

- **Surface:** Constitution and rights
  Governing question: What may not be traded away, even for aggregate performance?
  Failure if collapsed: Protected persons or values become optimization variables.
- **Surface:** Authority and jurisdiction
  Governing question: Who may decide what, for whom, where, and for how long?
  Failure if collapsed: A favorable outcome launders unauthorized power.
- **Surface:** Admissibility
  Governing question: Which proposed transitions may enter governed state?
  Failure if collapsed: Expected payoff substitutes for lawful permission.
- **Surface:** Incentive perception
  Governing question: What does an actor believe is costly, valuable, risky, or reachable?
  Failure if collapsed: Designer assumptions are mistaken for actor knowledge.
- **Surface:** Realized payoff and burden
  Governing question: What actually happened to each affected party?
  Failure if collapsed: Intended incentives substitute for observed effects.
- **Surface:** Evidence and observation
  Governing question: What can be measured, by whom, with what error and dependency?
  Failure if collapsed: Measurement is mistaken for the underlying property.
- **Surface:** Revision and contest
  Governing question: How can a bad model, rule, decision, or metric be challenged?
  Failure if collapsed: Optimization hardens its own assumptions into authority.

An action can be attractive yet inadmissible. An action can be costly yet constitutionally required. A right can remain binding even when violating it would increase aggregate performance. A lawful refusal may be strategically inconvenient. These are not exceptions to GIE; they define its boundary.

Let $Law(s, a, i)$ denote whether actor i may take action a in state s under the applicable jurisdiction. Let $V_i(s, a)$ denote an actor-relative valuation. Then:

$$V_i(s, a) > V_i(s, b)$$

does not imply:

$$Law(s, a, i) = ADMIT.$$

Incentive analysis operates inside or alongside a constitutional envelope. It does not create that envelope's authority.

3. A Governed Strategic Transition Model

A configured GIE study may be represented as:

$$G = \langle S, N, A, T, J, O, E, U, C, L, H \rangle,$$

where:

- S is the set of represented states;
- N is the set of actors and affected parties;
- $A_i(s)$ is actor i's represented action set at state s;
- T is the transition relation, including exogenous events;
- J is the jurisdiction and admissibility regime;
- O is the observation and evidence model;
- E is the external-effect model;
- U_i is actor i's declared or inferred valuation model;
- C_i is actor i's vector of costs and burdens;
- L is the learning, amendment, contest, and revision regime;
- H is the time horizon and discount or persistence model.

The admissible action set is:

$$A_i^{adm}(s) = \{a \in A_i(s) : J(s, a, i) = ADMIT\}.$$

`DEFER`, `REFUSE`,
and `GAP` remain distinct outcomes. A strategy is a policy over histories, not merely a single action. For path π and environment condition ω:

$$C_i(\pi \mid s_0, \omega, H) = \langle c_b, c_c, c_d, c_a, c_p, c_x, c_v, c_o, c_h \rangle_i ,$$

where the components provisionally denote binding, correction, deception, audit, participation, contest-or-exit, privacy, coordination, and displaced-harm costs.

These costs are actor-relative, role-relative, time-dependent, and uncertain. The same filing requirement may be trivial for an institution and prohibitive for an individual. A visibility rule may lower audit cost while raising privacy and retaliation cost. A correction path may exist formally while requiring expertise, time, language access, or money unavailable to the affected party.

No universal scalar **energy** follows from this vector. A study may define an actor-specific aggregation:

$$e_i(\pi) = \phi_i(C_i(\pi), U_i, H),$$

but ϕ_i must declare its weights, ordering rule, uncertainty, and authority. Different actors may rank the same paths differently. A constitutional analysis must preserve those differences rather than average them away automatically.

The model is intentionally incomplete. Unrepresented actors, hidden actions, changing preferences, power asymmetries, and model error are first-class limitations, not residual noise.

4. From Metaphor to Testable Topology

The source uses manifolds, gradients, attractors, basins, sinks, reservoirs, and ridges. These terms can be useful at three different levels:

1. **pedagogical metaphor**, used to make strategic structure intuitive;
2. **graph or state-space analysis**, where reachability, transition probability, recurrence, and barrier cost are measurable;
3. **mathematical topology or dynamical systems**, where a state space, neighborhood relation, metric or topology, dynamics, and stability criterion are explicitly defined.

The paper currently supports the first level and proposes a route into the second. It does not yet establish the third generally. A GIE study must state which level it uses.

- **Source term:** Corruption attractor
  Operational candidate: A region in which strategy profiles that increase governance divergence recur or persist after bounded perturbations.
  Evidence needed: State variables, divergence criterion, recurrence window, actor model, perturbation protocol.
  Defeater: The profile dissolves under ordinary correction or was an artifact of measurement.

- **Source term:** Correction basin
  Operational candidate: A set of states from which accepted counterevidence reaches lawful revision within bounded time and burden.
  Evidence needed: Seeded errors, admissible challenges, correction outcomes, distribution of burdens.
  Defeater: Valid challenges are suppressed, delayed beyond the bound, or corrected only for privileged actors.

- **Source term:** Semantic sink state
  Operational candidate: A state from which relevant meaning, authority, or lineage cannot be reconstructed sufficiently to decide lawful continuation.
  Evidence needed: Reconstruction task, required fields, independent verifier, failure threshold.
  Defeater: The state remains reconstructable from retained evidence.

- **Source term:** Perturbation reservoir
  Operational candidate: A bounded queue or state for unresolved anomalies that preserves custody without premature binding or erasure.
  Evidence needed: Intake, retention, prioritization, aging, escalation, and disposition receipts.
  Defeater: The reservoir becomes a silent graveyard or denial-of-service surface.

- **Source term:** Amendment gradient
  Operational candidate: The change in actor-relative burden and probability of success along a lawful amendment path.
  Evidence needed: Named path, steps, role costs, delay, success, and appeal data.
  Defeater: The gradient depends mainly on hidden discretion or inaccessible resources.

- **Source term:** Authority concentration ridge
  Operational candidate: A narrow set of actors, dependencies, or decisions through which many admissible paths must pass.
  Evidence needed: Authority graph, minimum controlling coalition, delegation and revocation paths.
  Defeater: Apparent concentration disappears when actual effect authority is traced.

Semantic Manifold Topology [SMT] supplies a language for thick typed relations and reachable continuations. GIE adds strategic actors, admissibility, observations, burdens, effects, and temporal behavior. Using topological vocabulary does not merge the two works or make every semantic neighborhood an optimization landscape.

5. The Cost Surface

5.1 Binding Cost

Binding cost is the actor-relative burden required for a proposal, claim, decision, or artifact to acquire a specified governed consequence. It must be indexed by the binding type and authority involved.

Too little binding friction can admit unsupported or unauthorized effects. Too much can exclude low-resource participants, freeze correction, or concentrate practical authority in actors able to pay the cost. The target is therefore not maximum binding cost. It is a justified, accessible, risk-proportionate path whose burden is measured across affected roles.

5.2 Correction Cost

Correction cost is the actor-relative burden required to identify, contest, revise, compensate for, or retire a binding. Correction cost includes more than the maintainer's implementation work. It includes discovery, standing, representation, delay, proof, retaliation risk, recovery, and irreversible loss.

Low formal amendment cost can still produce instability. High practical correction cost can preserve a formally corrigible but operationally captured system. Correction evidence should therefore report both the declared path and its realized accessibility.

5.3 Deception Cost

Deception cost is the burden of sustaining a specified misleading strategy against a stated observation and challenge regime. The term must not presuppose access to intention. In most studies, the observable object is a **deceptive or misleading strategy class**, not an actor's inner honesty.

Let π_D be a tested misleading strategy and π_R a tested reality-responsive strategy. For a declared scalarization only:

$$\Delta_i = e_i(\pi_D) - e_i(\pi_R).$$

If $\Delta_i > 0$ across the tested conditions, the evidence supports only this statement:

> Under this actor model, horizon, observation regime, and cost aggregation, the tested misleading strategy was more costly than the tested reality-responsive strategy.

It does not prove honest intent, global optimality, truth, legitimacy, or behavior outside the observed conditions.

5.4 Additional Cost Dimensions

The source's three dimensions are diagnostic anchors, not an exhaustive basis. Every study should also inspect:

- **audit cost**: burden of producing and independently checking evidence;
- **participation cost**: burden of entering, understanding, and using the process;
- **contest cost**: burden of challenging a decision without retaliation;
- **exit and substitution cost**: burden of leaving, forking, or changing providers;
- **privacy and disclosure cost**: personal exposure required for observability;
- **coordination cost**: burden of assembling witnesses, coalitions, or dependencies;
- **delay cost**: harm caused while a decision or correction remains unresolved;
- **harm-displacement cost**: burdens exported to actors or domains outside the measured boundary.

A design that increases deception cost by imposing universal surveillance may be strategically effective and constitutionally inadmissible. A design that lowers aggregate correction time by excluding difficult claims may improve the metric while worsening governance. GIE needs constitutional and distributional gates precisely because incentives alone cannot decide these tradeoffs.

6. Reality Coupling Without a Reality Oracle

The source distinguishes empirical, predictive, operational, intersubjective, and constraint coupling. This plural model should be retained.

For a study at time t, let:

$$R_t = \langle r_{emp}, r_{pred}, r_{op}, r_{inter}, r_{con} \rangle_t.$$

Each component needs a domain-specific observation contract. A system can score well on one component and fail another. Predictive accuracy can coexist with injustice. Operational survival can coexist with coercion. Intersubjective agreement can result from dependence, exclusion, or capture. Constraint compliance can preserve a harmful status quo.

Reality coupling therefore does not establish legitimacy. It describes whether selected external constraints can perturb selected governed beliefs or actions through a declared evidence path.

The phrase **external contradiction** also needs qualification. Contradictions do not enter governance by magic. A study must state:

- who may submit the observation;
- how the observation is authenticated without requiring excessive disclosure;

- which ontology and comparison rule make it contradictory;
- which authority can admit, contest, defer, or refuse it;
- how minority, novel, or unpopular evidence avoids automatic suppression;
- how malicious flooding and witness capture are handled;
- what remains preserved when the observation cannot yet be adjudicated.

A GIE design should prefer multiple independently controlled evidence paths where feasible. **Witness independence** is an empirical relationship among funding, infrastructure, governance, data, personnel, and control. It is not established by assigning different witness names.

7. Observability Is a Governed Scarce Resource

Observability determines which strategic distinctions the system can detect. It is bounded by attention, computation, access, privacy, latency, expertise, and institutional capacity. Adversaries may attack the observation layer through flooding, selective disclosure, synthetic evidence, coordination, delay, vocabulary manipulation, or capture.

Let O map an underlying path and environment to an evidence record:

$$O(\pi, \omega) \rightharpoonup y \mid PARTIAL \mid CONFLICT \mid GAP.$$

The same evidence record may be compatible with multiple underlying paths. GIE must therefore distinguish:

- underlying conduct;
- observable traces;
- admitted evidence;
- interpreted findings;
- binding decisions;
- external effects.

Observability collapse occurs when the declared decision distinctions can no longer be supported at the required error, latency, capacity, or independence bounds. It should be diagnosed by failed discrimination and reconstruction tests, not by a general sense of informational disorder.

More observation is not automatically better. Total observability can destroy privacy, dissent, experimentation, or safety. The objective is **minimum sufficient, purpose-bound, contestable observability**, coupled to retention and disclosure limits.

8. Ontology Laundering and Metric Capture

The source's strongest adversarial contribution is **ontology laundering**: altering the categories through which conduct is judged so that a strategy becomes admissible without becoming less harmful.

Ontology laundering can occur through:

- redefining evidence as completion of a procedure;
- redefining accountability as possession of a receipt;
- redefining safety as centralized permission;
- redefining dissent as attack or noise;
- redefining correction as reputational protection;
- narrowing the measured population until harms disappear from view.

The defense cannot be an eternally fixed vocabulary. Fixed vocabularies can preserve obsolete or oppressive categories. The required structure is governed semantic revision:

1. preserve the predecessor term, definition, scope, and evidence;
2. make the proposed semantic change explicit;
3. expose who benefits, who loses standing, and which decisions change;
4. admit counterexamples and minority interpretations;
5. test continuity and incompatibility across the change;
6. authorize the successor under a declared process;
7. retain contest, fork, and revision paths.

Metrics are part of the ontology. Once correction time, deception survivability, witness independence, or any other measure becomes a target, actors can optimize the measurement process rather than the underlying property. Every metric therefore needs an anti-Goodhart record: known proxies, gaming routes, omitted populations, uncertainty, and conditions under which the metric must lose standing.

9. The Governance Incentive Engineering Loop

GIE operates as a governed experimental and revision loop:

1. **Bound the domain.** Name the system, population, actors, affected nonparticipants, jurisdictions, effects, horizon, and excluded regions.
2. **Declare constitutional floors.** Identify rights, negative powers, authority limits, privacy constraints, and values that may not be optimized away.
3. **Model strategies and transitions.** Represent declared rules, actual mechanics, actor perceptions, hidden actions, evidence paths, effects, and burdens.
4. **State hypotheses.** Name the harmful and preferred strategy classes and the conditions under which their relative viability is expected to change.
5. **Generate CM1.** Enumerate admissible configurations, including centralized, expensive, exclusionary, surveillance-heavy, brittle, and otherwise undesirable ones.
6. **Select CM2.** Prefer configurations using explicit criteria for accessibility, burden distribution, privacy, contest, exit, correction, robustness, and evidence quality.

7. **Choose an Instance Configuration.** Bind exact actors, policies, thresholds, measures, authorities, datasets, witnesses, software, environments, and recovery rules.
8. **Realize under separate authority.** Implement only within the authorized domain and effect boundary.
9. **Evaluate adversarially.** Test ordinary use, low-resource use, collusion, capture, mimicry, flooding, exit, sabotage, observation failure, burden shifting, and model misspecification.
10. **Preserve evidence and counterevidence.** Record successes, failures, unknowns, dissent, harms, and invalidated assumptions.
11. **Revise or retire.** Update the model, configuration, rule, metric, or system; preserve predecessor lineage and changed claim ceilings.

This loop is related to the Governed Knowledge Loop, but it is not automatically identical to it. GIE focuses on strategic topology and intervention evaluation. The Governed Knowledge Loop focuses on the lifecycle of epistemic artifacts. A configured program may use both.

10. Worked Example: A Governed Technical-Correction Process

Consider a low-consequence repository or research corpus in which participants report technical errors and maintainers decide whether to correct a canonical artifact.

10.1 Declared Process

The process states that anyone may submit an error report, maintainers review evidence, accepted errors produce a revision, and every decision is receipted.

Rule inspection alone suggests corrigibility. Operational inspection may reveal that:

- reports require insider vocabulary;
- only maintainers can reproduce the build;
- rejection reasons are generic;
- repeat submitters face retaliation or exhaustion;
- accepted reports wait indefinitely;
- a receipt proves closure but not correction;
- maintainers and witnesses depend on the same infrastructure and authority.

10.2 Model

Actors include reporter, maintainer, independent reproducer, affected user, and project authority. Strategies include valid report, noisy report, strategic denial, compliant closure, genuine correction, delay, forum exit, and outside disclosure.

The constitutional floor includes:

- no retaliation for good-faith reports;
- no requirement to disclose irrelevant personal data;
- preserved source and report custody;
- visible reasons and a contest path;
- no silent mutation of the canonical artifact;
- claim scope limited to the evidence actually reproduced.

The study compares configurations such as:

- maintainer-only reproduction;
- independent reproduction for high-impact claims;
- public evidence by default;
- privacy-preserving sealed evidence with later disclosure rules;
- fixed response windows with `DEFER` and `GAP` states;
- appeal to a separate reviewer;
- provider-exit and fork support.

10.3 Measurement

For seeded and naturally occurring reports, measure by actor and severity:

- time and work to submit;
- probability of reaching substantive review;
- false acceptance and false rejection;
- time from accepted evidence to actual correction;
- ability to reconstruct the decision and changed artifact;
- retaliation, privacy, and disclosure burden;
- minimum coalition able to block correction;
- persistence of a known false canonical statement;
- difference between closure receipts and material outcomes.

A configuration improves the tested correction basin only if valid reports reach lawful correction with bounded and equitably distributed burdens while false reports, flooding, and privacy attacks remain manageable. Faster closure alone is not improvement. More receipts alone are not correction. Greater central control alone is not witness independence.

This example shows the value of GIE without claiming that one metric or configuration generalizes to institutions at civilization scale.

11. Measurement Contracts

Every metric must declare its unit, population, horizon, sampling rule, observation procedure, uncertainty, dependencies, gaming routes, and falsifier.

- **Metric candidate:** Correction cost
  Minimum operational contract: Distribution of time, money, expertise, procedural steps, risk, and irreversible harm from discovery through verified correction.

Required companion measures: False correction, rejected valid challenge, burden by role, delay harm.

- **Metric candidate:** Deception survivability
  Minimum operational contract: Duration or transition count for a seeded misleading strategy to remain binding under a named audit regime.
  Required companion measures: Detection error, untested strategy classes, audit capacity, privacy burden.
- **Metric candidate:** Perturbation permeability
  Minimum operational contract: Fraction of validated external challenges reaching substantive adjudication within a bound.
  Required companion measures: Flooding load, false admission, false suppression, reporter accessibility.
- **Metric candidate:** Semantic viscosity
  Minimum operational contract: Work and time required to revise an accepted semantic binding while preserving lineage.
  Required companion measures: Continuity loss, instability, affected-party burden, rollback and forkability.
- **Metric candidate:** Witness independence
  Minimum operational contract: Dependency graph across control, funding, data, infrastructure, personnel, and authority.
  Required companion measures: Common-mode failure, collusion tests, minimum controlling coalition.
- **Metric candidate:** Authority concentration
  Minimum operational contract: Smallest actor set able to admit, block, reinterpret, or effect high-impact transitions.
  Required companion measures: Delegation, revocation, hidden vetoes, practical accessibility.
- **Metric candidate:** Drift accumulation
  Minimum operational contract: Versioned semantic differences from a declared baseline, weighted only by an explicit significance rule.
  Required companion measures: Legitimate amendment, baseline error, contested meanings, model uncertainty.
- **Metric candidate:** Observability integrity
  Minimum operational contract: Ability to distinguish specified strategy classes under bounded noise, load, latency, and disclosure.
  Required companion measures: Privacy, selective visibility, false positives, false negatives, capacity exhaustion.

No row is yet a universal metric. A publication-ready empirical paper would need a domain, instrument, dataset, study protocol, statistical model, and independent replication plan.

12. Corpus Placement and Division of Labor

GIE is most useful when its relationship to adjacent Ableman work is explicit.

- **Semantic Manifold Topology** represents thick typed relations, contextual neighborhoods, and reachable continuations [SMT]. GIE adds actors, law, cost, observation, strategy, and temporal stability.
- **Governance-Aware Optimization** places a governance transform between proposed numeric updates and parameter commitment [GAO]. It is a specific machine-learning realization profile, not the whole GIE discipline.
- **Constitutional Epistemology Under Adversarial Semantic Pressure** supplies a threat model for governance mimicry, witness pressure, and semantic attack [CEASP]. GIE asks how those strategies change under specified mechanisms and costs.
- **Admissibility Engineering** moves constraints from prose toward artifact-layer enforcement and examines saturation gaps [AE]. GIE examines the strategic behavior induced by those constraint surfaces.
- **Constitutional Computing** separates law, authority, admissibility, effect, evidence, and revision [CC]. GIE evaluates the strategy landscape inside and around a configured constitutional system.
- **Sovereign Software** supplies user-facing inspectability, substitutability, exit, and renegotiation concerns [SS]. These are constitutional and distributional constraints on incentive design.
- **The Ableman Prime Directive** foregrounds detectability, refusal, exit, and renegotiation of bindings [APD]. GIE must not raise deception cost by destroying these negative powers.
- **Reasoning by Way of Receipts** and **The Three Hash Rule** distinguish claims from exact evidence and lineage [RWR] [THR]. GIE uses receipts as evidence inputs, never as substitutes for material outcomes.
- **Constraint-Bearing Composability** asks whether obligations survive composition [CBC]. GIE must test whether local incentive improvements survive interfaces, adapters, institutions, and downstream effects.
- **The Constitutional Observability Platform** supplies an adjacent observability and evidence surface [COP]. GIE supplies adversarial and distributional requirements for what that surface must detect without turning observation into ambient surveillance.
- **The Governed Knowledge Loop** governs evolving epistemic artifacts and contradiction intake [GKL]. GIE results, models, disputes, and revisions can move through that loop without becoming truth by registration.

The current meaning-base name is **Governed Meaning Base (GMB)** in the historical lineage `CCF -> CMB -> GMB`. GMB can store the exact GIE specification, cost model, actor model, evidence, counterevidence, intervention, metric contract, dispute, standing, and revision history. It is a knowledge and meaning substrate, not an objective-reality oracle, incentive authority, or institutional sovereign. Hyperpedia may project this material pedagogically while preserving source custody and standing.

13. Authorial Compression and External Comparison

GIE may be a distinct corpus-level discipline, an integrative method, or a profile assembled from existing machinery. The current internal differential is:

1. constitutional constraints are non-reducible to utility or payoff;
2. costs are typed, actor-relative, distributional, and evidence-bound;
3. semantic and ontological change are part of the strategic surface;
4. observation, receipts, effects, and claim ceilings remain distinct;
5. interventions move through CM1, CM2, Instance Configuration, evidence, and revision;
6. negative power, contest, privacy, pluralism, exit, and burden shifting constrain preference;
7. the GMB retains the complete model and its disputes without granting it truth or authority.

If these obligations do not produce different analyses, experiments, or design decisions from adjacent work, **Governance Incentive Engineering** should be compressed into a profile or method rather than retained as a foundationally new system.

The paper requires external comparison with mechanism design, institutional economics, game theory, security economics, control theory, adversarial machine learning, socio-technical systems, constitutional design, organizational behavior, and dynamical-systems analysis. No such external research was conducted in this pass. Novelty, priority, and disciplinary distinctness remain publication blockers.

Strategic-Governance Analysis and Intervention Method

Across the completed P1 corpus, Governance Incentive Engineering is best retained as the **integrative method for modeling, testing, and revising the strategy landscapes produced by constitutional arrangements**. Its distinct contribution is the actor-relative, distributional connection among law, admissibility, observation, effects, costs, contest, exit, and revision. It is not a replacement constitution, a utility theory of legitimacy, or a general promise that governance can make adversaries honest.

- **Adjacent work:** *Semantic Field Analysis*
  Supplies: Profile-relative graph diagnostics, tension, paths, and sensitivity.
  GIE boundary: GIE adds actors, strategies, burdens, effects, and interventions to a bounded governed field.

- **Adjacent work:** *Admissibility Engineering*
  Supplies: Constraint surfaces and artifact-layer refusal.
  GIE boundary: GIE measures the strategies and burden distributions induced by those surfaces.

- **Adjacent work:** *Constitutional Epistemology Under Adversarial Semantic Pressure*
  Supplies: Adversarial evidence, witness, interpretation, and semantic threats.
  GIE boundary: GIE converts selected threat classes into testable strategic hypotheses.
- **Adjacent work:** *Agentic Interface Surfaces*
  Supplies: Action, effect, evidence, feedback, and recognition surfaces.
  GIE boundary: GIE evaluates how those surfaces change actor or agent behavior without reducing governance to reward.
- **Adjacent work:** *The Constitutional Observability Platform*
  Supplies: Heterogeneous observation and result visibility.
  GIE boundary: GIE specifies minimum sufficient observability and tests capture, privacy burden, and common-mode failure.

A GIE intervention claim should bind:

$$\begin{aligned} I = (&constitution, domain, actors, affected_parties,\\ &strategies, observations, effects, cost_vectors,\\ &baseline, intervention, distribution, counterfactual,\\ &failures, revision, claim_ceiling) \end{aligned}.$$

An intervention succeeds only relative to named harmful and preferred strategy classes while preserving constitutional floors. Aggregate improvement cannot hide exclusion, retaliation, privacy loss, displaced harm, inaccessible correction, capture, or provider lock-in. Where an incentive hypothesis conflicts with a right or authority boundary, the constitutional boundary governs.

Every metric is also a strategic surface. Measurement contracts must expose proxies, omitted populations, dependencies, gaming paths, capacity limits, and who can change the metric. Clean compliance, more receipts, or faster closure are insufficient unless independently bound to the intended external outcome.

The universal GSM can type
`INDUCES_STRATEGY_LANDSCAPE`, `BEARS_COST_AS_ACTOR`,
`AFFECTED_WITHOUT_AGENCY`, `CONSTRAINED_BY_CONSTITUTIONAL_FLOOR`,
`OBSERVED_UNDER_MEASUREMENT_CONTRACT`, `ALTERED_BY_INTERVENTION`,
`DISPLACES_HARM_TO`,
`CAPTURES_METRIC`, `PRESERVES_CONTEST_AND_EXIT`,
and
`REVISED_AFTER_STRATEGIC_FAILURE`. These relations establish the method's differential without declaring a new sovereign system.

This integration establishes four editorial commitments: classify GIE as an integrative strategic-governance method; adopt the intervention-claim envelope; require distributional, counterfactual, anti-gaming, and affected-party analysis;

and make constitutional floors, refusal, contest, privacy, pluralism, and exit non-optimizable constraints.

Incentives, Lifecycle Mutation, and Custody of Intervention

The fixed-seed comparison with the international-trade UGK profile, Admissibility Theory of Consciousness, the Governed Knowledge Loop, and Work Custody clarifies the operational life of a GIE intervention:

$$\begin{aligned} I^* = (&governed_state, actors, strategies, measurement,\\ &proposal, custodian, admission, intervention, effects,.\\ &distribution, monitoring, reconsideration, revision) \end{aligned}$$

The trade profile supplies a concrete multi-actor domain where handoff rules, observation, delay, false admission, false refusal, and appeal can redistribute cost and strategic opportunity. Its title and publication identity remain unresolved, so it is a provisional application case rather than adopted proof of GIE.

The Governed Knowledge Loop can preserve the baseline, analysis, intervention proposal, evidence, mutation decision, successor, and rollback. Work Custody names who has the next-move duty to investigate, implement an authorized test, monitor consequences, respond to affected parties, or return the work. Custody does not confer mutation authority or make the custodian's preferred metric legitimate.

Admissibility Theory of Consciousness has no direct evidentiary relation to this intervention method. Shared language about viability, continuation, or reflexivity must not be used to infer consciousness, moral status, or strategic-governance validity. The nonrelation remains explicit.

The universal GSM can type
`PROPOSES_INCENTIVE_INTERVENTION`,
`ASSIGNS_INTERVENTION_CUSTODY`, `ADMITS_GOVERNED_MUTATION`,
`MONITORS_DISTRIBUTIONAL_EFFECT`, `RETURNS_INTERVENTION_FOR_RECONSIDERATION`,
`ROLLS_BACK_STRATEGIC_FAILURE`, and
`HAS_NO_EVIDENTIARY_RELATION_TO_ATC`.

This randomized review establishes four editorial commitments: adopt the intervention lifecycle; assign explicit custody without authority escalation; use trade only as an identity-held application profile; and retain the nonrelation to consciousness theory.

Constitutional Floors Around Strategic Intervention

Five downstream uses apply GIE to work custody, ATC comparison, CCOO, the knowledge loop, and international-trade authority chains. They confirm GIE as a method for analyzing and testing strategic pressure around governed

transitions. It is not a replacement for a constitution, a reality oracle, a license to manipulate participants, or proof that observed behavior reveals motives.

$$
\begin{aligned}
G^* = (&subject, actors, objectives, constitutional_floors,\\
&information, authority, incentives, costs, proxies,\\
&observation, baseline, intervention, affected_parties,.\\
&outcomes, distribution, gaming, appeal, rollback,\\
&revision)
\end{aligned}
$$

Mandatory rights, safety constraints, authority boundaries, privacy, due process, and refusal remain noncompensatory floors. Incentive analysis begins only inside the admitted region. Binding, correction, deception, delay, exit, coordination, observation, and contest costs should be reported by actor and distribution rather than collapsed into one objective function.

Observation changes incentives and can reward legibility rather than truth. Proxy and ontology drift, strategic disclosure, collusion, selection effects, and burden shifting require countertests. Interventions need a baseline, causal theory, affected-party consultation, bounded authority, reversible trial where possible, distributional evidence, stopping rule, appeal, and rollback. A successful local metric does not establish institutional legitimacy or general welfare.

The universal GSM can type
`MODELS_GOVERNANCE_INCENTIVE`,
`PRESERVES_CONSTITUTIONAL_FLOOR`, `DISTRIBUTES_INTERVENTION_COST`,
`GAMES_GOVERNANCE_PROXY`,
`OBSERVES_STRATEGIC_RESPONSE`, `APPEALS_GIE_INTERVENTION`, `ROLLS_BACK_INCENTIVE_CHANGE`,
and
`DOES_NOT_INFER_MOTIVE`.

This review establishes four editorial commitments: adopt the strategic-intervention record; keep constitutional floors noncompensatory; require actor-distributed costs and proxy countertests; and gate interventions through consultation, authority, reversible trial, stopping, appeal, rollback, and revision.

14. Counterevidence and Failure Modes

Evidence against a GIE model includes more than failure to change average behavior. The framework should be revised or refused when:

- a supposedly preferred strategy is inaccessible to low-resource actors;
- the intervention moves cost from powerful actors to affected outsiders;
- observed compliance rises while unobserved harm, mimicry, or retaliation rises;
- the witness layer shares control, data, funding, or infrastructure with the governed actor;

- participants optimize labels, receipts, queues, or metrics instead of the intended property;
- lawful dissent, minority evidence, novelty, or ambiguity is classified as noise or attack;
- privacy loss or compelled disclosure exceeds the justified observation need;
- the mechanism centralizes interpretation or effect authority;
- actors exit, sabotage, collude, or move conduct to unobserved channels;
- non-economic motives make the assumed payoff model misleading;
- delayed or irreversible harms occur before correction can operate;
- multiple principals impose incompatible objectives or jurisdictions;
- local improvement exports harm to another system, population, or time horizon;
- the topology depends on hidden discretion rather than declared transitions;
- a compliant strategy is mistaken for honest belief or valid consent;
- the model cannot reconstruct important effects from its evidence;
- intervention overhead destroys the activity it was meant to govern;
- a simpler capability, access-control, audit, or correction mechanism provides the same assurance;
- an adversary changes the rules, observation process, actor identities, or boundary itself.

The framework should preserve null results. If an intervention raises procedural cost without changing the harmful strategy, that finding is more valuable than a narrative claim that the topology improved.

15. Evaluation and Conformance Program

A bounded GIE study should provide:

1. an exact domain and system boundary;
2. actor and affected-party registry with uncertainty and omitted-population analysis;
3. constitutional floors, authorities, jurisdictions, rights, and non-optimizable constraints;
4. state, action, transition, observation, effect, learning, and horizon definitions;
5. typed cost vectors by actor and role;
6. declared strategy classes, including mimicry, capture, collusion, exit, and sabotage;
7. metric contracts and anti-gaming analysis;
8. CM1 configuration enumeration and CM2 preference criteria;
9. an exact Instance Configuration and change record;
10. baseline, counterfactual, and intervention protocols;
11. seeded-error, seeded-deception, load, privacy, and accessibility tests;
12. false-positive, false-negative, and observability-collapse analysis;
13. burden and benefit distributions rather than aggregate results alone;
14. independent evidence paths and dependency analysis;

15. receipts binding claims to exact artifacts, models, configurations, and results;
16. counterevidence, dissent, unknowns, and invalidated assumptions;
17. revision, rollback, compensation, exit, and retirement paths;
18. claim-ceiling tests that reject conclusions broader than the evidence.

Passing such a program would support only the named instance and study scope. It would not prove GIE universally, establish truth, authorize the constitution, or transfer standing to another domain.

16. Realization Funnel

Stage	GIE role
Possibility	Explore actors, rules, strategies, semantic models, cost dimensions, observations, interventions, beneficial outcomes, harmful regimes, coercive designs, capture routes, and unknowns.
Specification	Define the governed strategic transition model, constitutional floors, terms, hypotheses, metrics, evidence, refusal conditions, and nonclaims.
CM1	Enumerate configurations that satisfy the specification, including centralized, exclusionary, costly, surveillance-heavy, fragile, or strategically perverse variants.
CM2	Prefer configurations with bounded authority, accessible correction, practical negative power, privacy, pluralism, burden fairness, substitution, robust evidence, and low harm displacement.
Instance Configuration	Bind the exact domain, actors, roles, law, authorities, thresholds, metrics, datasets, witnesses, software, infrastructure, horizon, recovery, and tests.
Implementation Realization	Build or alter the authorized technical and institutional mechanisms for that exact configuration; this paper grants no such authority.

Stage	GIE role
Evidence	Compare baseline, counterfactual, and intervention outcomes; preserve distributional effects, failures, unknowns, counterevidence, and exact configuration lineage.
Revision	Change or retire the model, ontology, metric, mechanism, law, configuration, or claim when evidence, harms, authority, environment, or affected populations change.

GIE primarily supplies a Specification-through-Evidence method. It does not collapse Specification into Implementation or an attractive CM1 configuration into a preferred CM2 choice.

Governance Is an Optimization Landscape, Not a Behavioral Oracle

Public orientation

Governance Incentive Engineering studies how rules, evidence, delays, rewards, sanctions, and appeal routes shape the strategies available to participants. A constitution can be formally correct while making evasion, delay, superficial compliance, or capture the locally rational move. The framework proposes diagnosis and intervention; it does not predict human behavior with certainty.

Adversarial tests

1. **Metric-Goodhart attack.** A measure becomes a target and loses connection to the protected purpose. Metrics need adversarial monitoring and plural evidence.
2. **Compliance-theater attack.** Participants optimize receipts or visible checkpoints while harmful behavior moves outside the observed boundary.
3. **Power-asymmetry attack.** The same incentive has different effects on actors with unequal resources, exit options, information, or appeal access. Distributional effects must be explicit.
4. **Manipulation attack.** Designers use governance mechanisms to steer people without meaningful notice or consent. Legitimacy and participant agency are independent of effectiveness.
5. **Crowding-out attack.** External incentives weaken intrinsic motivation, trust, or cooperation. Interventions need counterfactual and longitudinal evaluation.
6. **Equilibrium-shift attack.** A local repair creates a worse system-level strategy or coalition. Analysis must include adaptation, collusion, and

second-order effects.

7. **Model-misspecification attack.** Actors, utilities, information, or constraints differ from the model. Predictions must carry assumptions and uncertainty.
8. **Intervention-custody attack.** A policy changes without preserving who authorized it, what evidence justified it, or which outcomes followed. Governance experiments need versioned lineage and rollback.

Claim and evidence ceiling

The paper establishes an analysis vocabulary, measurement contracts, and a governed intervention lifecycle. It does not establish universal incentive laws, reliable prediction of persons or institutions, ethical acceptability, or effectiveness of a specific intervention.

Claims about economics, psychology, organizations, law, public policy, behavior, or social welfare require current external research, domain expertise, participant analysis, and empirical evaluation. Internal corpus consistency cannot validate behavioral assumptions. No external research was added in this pass.

Reader-facing consequence

The useful question is not only "what does the rule require?" but "what behavior does this arrangement make easiest, safest, most rewarding, most delayable, or most invisible - and for whom?"

17. Research Agenda and Open Questions

1. Decide whether GIE should remain a distinct discipline, become an integrative method within Constitutional Computing, or be presented as a profile spanning existing systems.
2. Accept or revise the separation among constitutional law, admissibility, incentives, actor perception, realized burden, evidence, and effect.
3. Accept or revise the governed strategic transition tuple and typed cost vector.
4. Decide whether **energy** remains the preferred pedagogical term after restricting it to an explicitly declared scalarization.
5. Define the minimum operational criteria for corruption attractor, correction basin, semantic sink, perturbation reservoir, amendment gradient, and authority concentration ridge.
6. Decide whether the five reality-coupling dimensions are sufficient and how normative legitimacy remains separate.
7. Select one or more low-consequence empirical domains for later study under separate implementation and research authority.
8. Define the constitutional floors for privacy, dissent, contest, exit, pluralism, accessibility, and burden distribution.

9. Decide the relation among GIE, Semantic Manifold Topology, Governance-Aware Optimization, Constitutional Epistemology, Admissibility Engineering, and the Governed Knowledge Loop without automatic identity merger.
10. Authorize external research before any Evidence-Readiness, novelty, priority, or disciplinary-comparison claim.

18. Conclusion

Governance does more than declare rules. It changes what actors can attempt, what they can observe, which strategies remain viable, who bears the burden of proof, how correction travels, and where power accumulates. Governance Incentive Engineering names the disciplined study of those induced strategic conditions.

Its strongest formulation is not that a sufficiently curved governance manifold makes adversaries honest. Honesty is not directly observable, actors differ, costs are distributed, models are incomplete, and adversaries can alter the game. The defensible result is conditional:

> Within a declared constitutional boundary, actor model, observation regime, horizon, and cost contract, a governed intervention may make specified harmful strategies less viable while preserving specified preferred strategies, rights, and correction paths.

That result must be demonstrated rather than inferred from rules, receipts, or compliant appearance. It must survive low-resource use, witness dependency, metric gaming, privacy constraints, dissent, exit, collusion, capture, observation failure, and displaced harm. It must remain revisable when its ontology or model is wrong.

GIE therefore complements constitutional governance without replacing it. Constitution establishes the non-reducible envelope. Admissibility governs entry into consequential state. Incentive analysis examines the strategy landscape. Evidence tests the modeled consequences. The GMB preserves the full argument and its counterevidence. Revision keeps the design from converting its own temporary measurements into permanent authority.

Corpus-Wide Reference Positioning

This paper argues that declared rules do not determine behavior by themselves. Its nearest external comparison fields are governance, markets, provenance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance,

institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Optimization can make a chosen objective efficient while preserving a bad objective, missing distributional harms, strategic behavior, or inadmissible states. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E2**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E3**] Lawrence Lessig. *Code: Version 2.0*. Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

[**E4**] OECD. *Recommendation of the Council on Artificial Intelligence*. OECD/LEGAL/0449, adopted 2019 and updated 2024. legalinstruments.oecd.org source **Relationship:** Adjacent policy standard for accountability, traceability, transparency, safety, and human-centered values.

[**E5**] Harry Markowitz. "Portfolio Selection." *The Journal of Finance* 7(1), 1952, pp. 77-91. DOI: 10.1111/j.1540-6261.1952.tb01525.x. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x **Relationship:** Predecessor risk-return optimization model; included to distinguish scalar optimization from governance-aware admissibility.

[**E6**] U.S. Department of the Treasury. *Tax Expenditures*. Office of Tax Analysis. home.treasury.gov source **Relationship:** Primary official baseline for the definition and estimation limits of U.S. federal tax expenditures.

[**E7**] W3C. *PROV-O: The PROV Ontology*. W3C Recommendation, 30 April 2013. https://www.w3.org/TR/prov-o/ **Relationship:** Adjacent provenance standard; it models entities, activities, and agents, but does not by itself establish the paper's stronger authority or standing claims.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Constitutional Governance*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-79353AF163A4B0F1`;
source SHA-256
`FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Theory*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D2D97C4042FD281F`;
source SHA-256
`FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governance Engineering*. Ableman Corpus current edition; work identity `PAPER.PUB.18749870`;
source SHA-256
`372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Universal Governance Kernel (UGK).* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-1F9BFB2BF8370226`;
source SHA-256
`AA5106EA71E5DF120E141435C280DFC532715CA5DE0C75B8862BB7784ECCFACD`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Legitimacy Without Self-Ratification.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-EF2382F9397335AC`;
source SHA-256
`0186A39C808F8FDDB5C8F42384DF1A26678928434BD50A0EAE8D4947390F68A2`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Constitutional Markets.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-1D49CCBB6E47B403`;
source SHA-256
`BB97EAC6F1861C2ED17DC0E0B75D47B04716E7FF3E3AC2A9B2F6CABCC67A5250`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *ACG for Quants.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-8A19CB4C1BD55BCA`;
source SHA-256
`2FEA83087B8DA79472BBA449C9D5B252B6DBA67C2EAF0DF41C206B1B0EEEF254`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Ableman Corpus references [**GIE-SOURCE**] Adam Ableman. *Governance Incentive Engineering.* Exact author source SHA-256
`E620B9AE4E8DCCB10702BD1E3ACDC494834B207D48887A36366E3515DA64C70C`;
unpublished source revised here as an AI-assisted integrated edition.

[**SMT**] Adam Ableman. *Semantic Manifold Topology: A Simple Worked Example - Conversation.* Exact reviewed Markdown source SHA-256
`BAA41D5FCC820805F23676EED9B1542E0DC8744F7D7A145D21F4BE99A0D99BC9`;
supplied Zenodo title match, with exact publication identity retained separately.

[**GAO**] Adam Ableman. *Governance-Aware Optimization.* Exact reviewed source SHA-256
`3BF121672FC1509DDD6546856C6A5E8378E30A90ED32A81CA132528D9C04081F`;
supplied Zenodo title match, with exact publication identity retained separately.

[**CEASP**] Adam Ableman. *Constitutional Epistemology Under Adversarial*

Semantic Pressure. Exact reviewed source SHA-256 `909F67AD82D8D8C6119FFEABCC8F885EF8C0A095DE79BFA29A65FDB1DC9E64C3`; separately custodied reviewed source.

[**AE**] Adam Ableman. *Admissibility Engineering*. Exact reviewed source SHA-256 `FA80D5895A569839958193735B754BE62EE20E1B399A06CB3BA959614DEBF7A3`; separately custodied reviewed source.

[**CC**] Adam Ableman. *Constitutional Computing*. Exact reviewed source SHA-256 `A1EF0D1F7B5DF84A107E986421EAA95827AEDCB801B7DA63F72BEECA6047BE8D`; supplied Zenodo title match, with exact publication identity retained separately.

[**SS**] Adam Ableman. *Sovereign Software*. Exact reviewed source SHA-256 `D8A2707865C6636485A564546D91F2CCF45AAC80161B94AC56A3DA5CF2EB2C13`; separately custodied reviewed source.

[**APD**] Adam Ableman. *Turing Completeness and the Ableman Prime Directive*. Exact reviewed source SHA-256 `80F4FCB00B214D22BF0E0A2D67E726DBA4410BAA751300A24DAAD2610E6C395E`; separately custodied reviewed source.

[**RWR**] Adam Ableman. *Reasoning by Way of Receipts*. Exact reviewed source SHA-256 `8D884B96597EA6303A324173A7CE2D8C735A952E91FBE4C9EB6509FA6D6B195F`; separately custodied reviewed source.

[**THR**] Adam Ableman. 2026. *The Three Hash Rule*. Zenodo. DOI: `10.5281/zenodo.20788617`. Exact reviewed Markdown source SHA-256 `03F60317CC9AFB36DDBCF9BF1C6D185611666CC4D9AB290BD4BCC17DF4579B7C`.

[**CBC**] Adam Ableman. *Constraint-Bearing Composability*. Exact reviewed source SHA-256 `BEEA0B56DBF7DC5274307E95E08AF0576A9F91A729C1962765361ADA34488B62`; supplied Zenodo title match, with exact publication identity retained separately.

[**COP**] Adam Ableman. *The Constitutional Observability Platform*. Exact reviewed source SHA-256 `15CBE49A23A34F881A06EAB32AAFD95E9468EC0B7C19F47A795A34C854DF9E84`; separately custodied reviewed source.

[**GKL**] Adam Ableman. *The Governed Knowledge Loop*. Exact reviewed source SHA-256 `C8CF0E10ACD0A1CCD91789363F1B28E1D7E8DC1865396C27DAAB6A40C6D32C7C`; separately custodied reviewed source.

[**GMB-HISTORICAL**] Adam Ableman. *Constitutional/Crystallized Meaning Base*. Exact reviewed source SHA-256 `8680F801CDCA0B4C81F7751A788106454D34923993B4E7BBFF68F35A83ED0DA0`; historical CMB-phase source in the author-confirmed

`CCF -> CMB -> GMB` terminology lineage.

No external research was conducted in this pass. External literature review, current primary-source verification, and conventional bibliography are required for a future evidence-qualified revision readiness or novelty claims.

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *The Three Hash Rule.* Zenodo, 2026. doi:10.5281/zenodo.20788617

Adam Ableman *Semantic Manifold Topology.* Zenodo, 2026. doi:10.5281/zenodo.20257244

Adam Ableman *A Theory of Agentic Observation.* Zenodo, 2026. doi:10.5281/zenodo.20437813

Adam Ableman *CARA: Constrained Artificial Reasoning Architecture.* Zenodo, 2025. doi:10.5281/zenodo.18099715

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Adam Ableman *Governed Semantic Compilation: Lawful Continuation, Refusal Semantics, and Epistemic Promotion in Large Language Models.* Zenodo, 2026. doi:10.5281/zenodo.18453681

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus

License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.23847865